Sri Lanka Institute of Information Technology

IT3010 – Network Design & Management (NDM)

Practical 01 - Report

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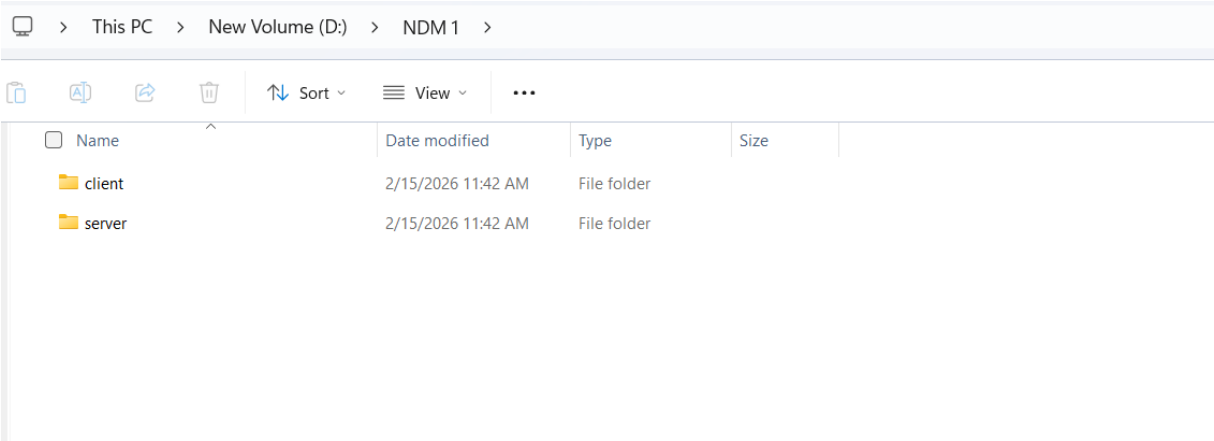
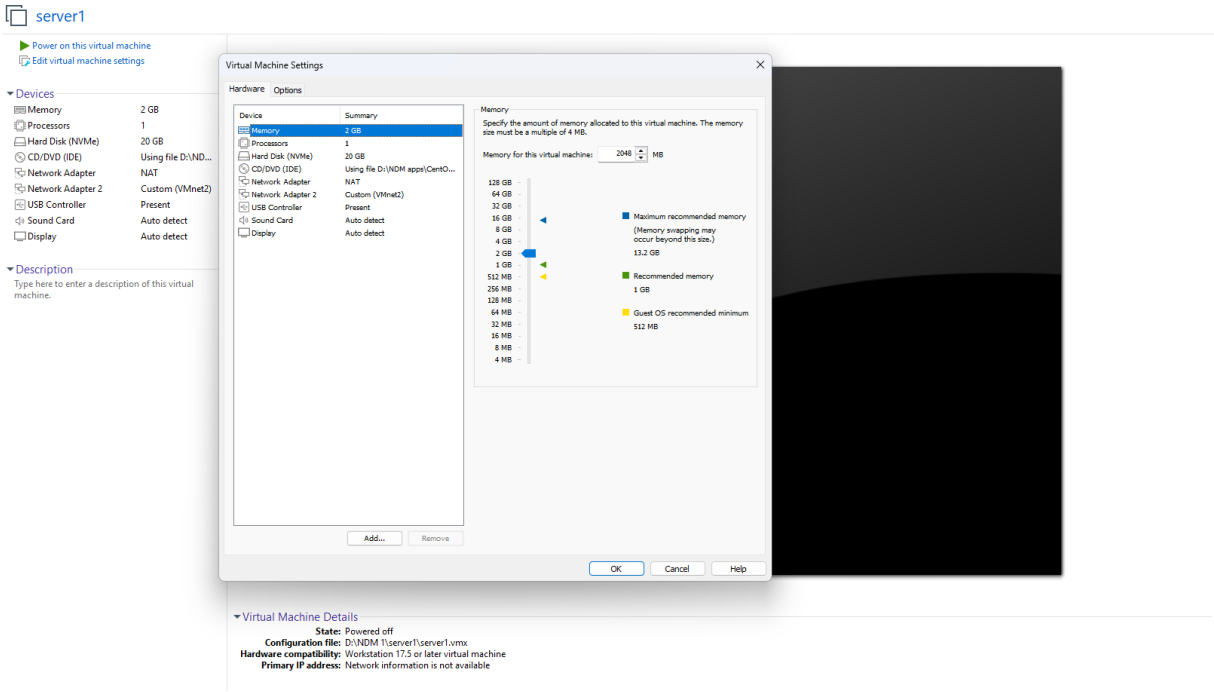
Contribution

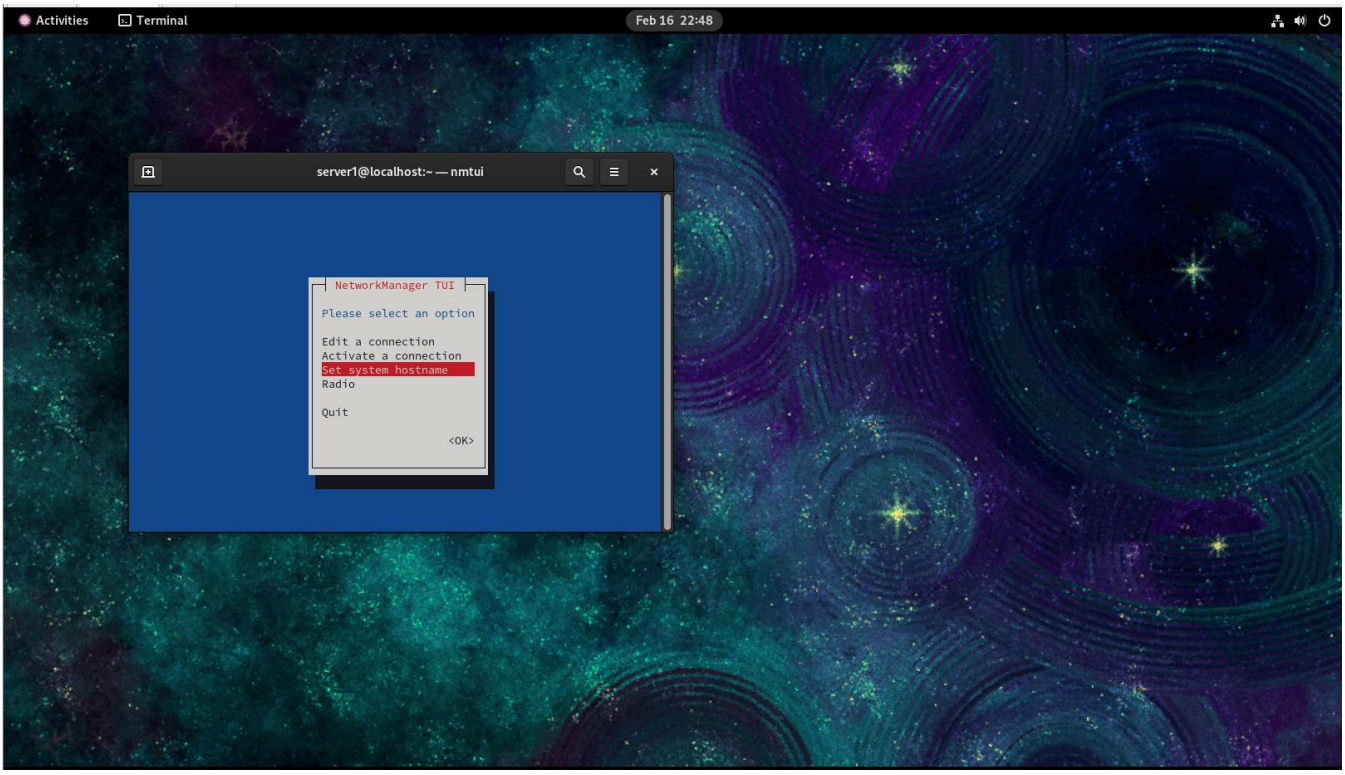
Name	Registration Number	Signature
PRIYAWANTHA M G C	IT23264748	

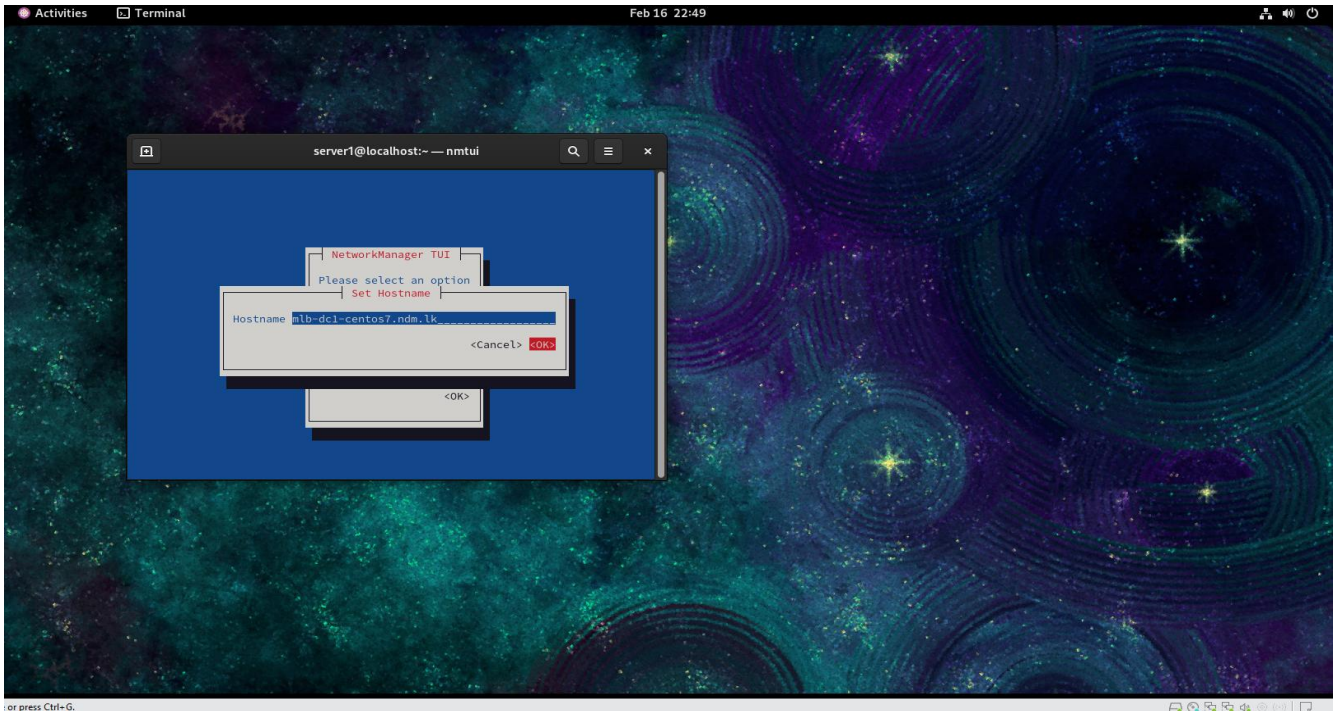
1.Create another two sub-directories called NDM-Server and NDM-Client.



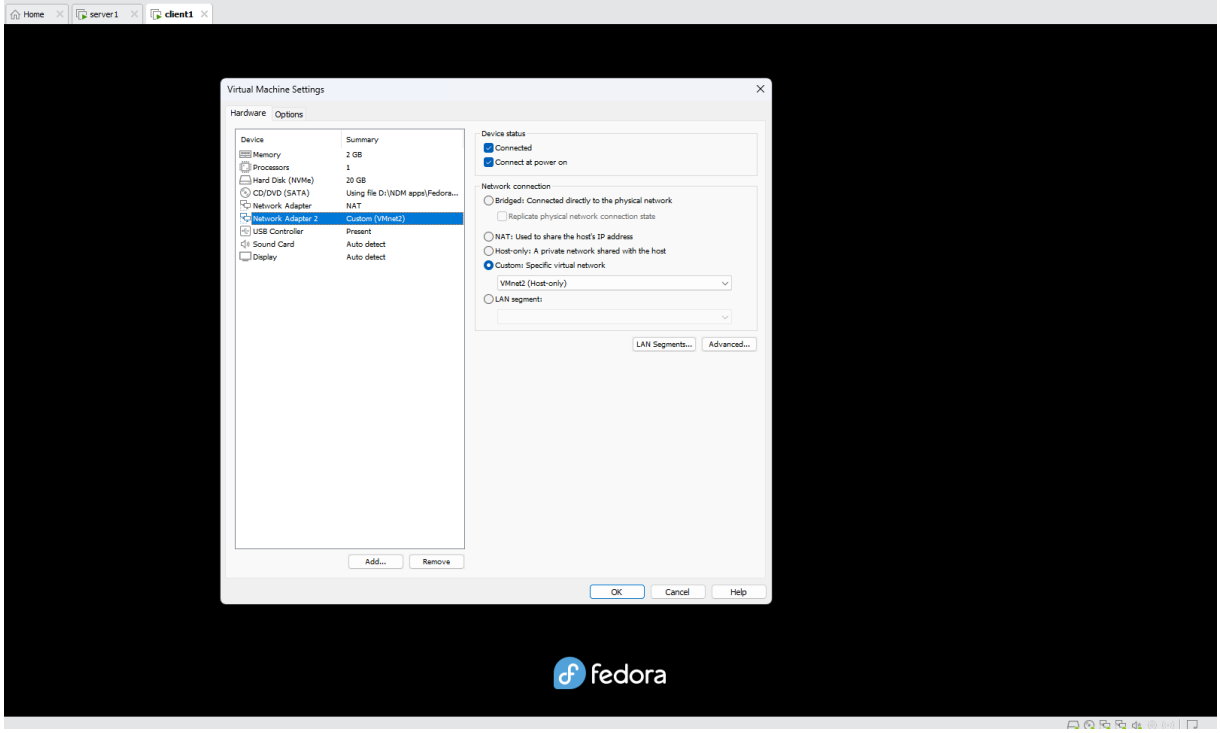
2.Creating the Server – Specifications



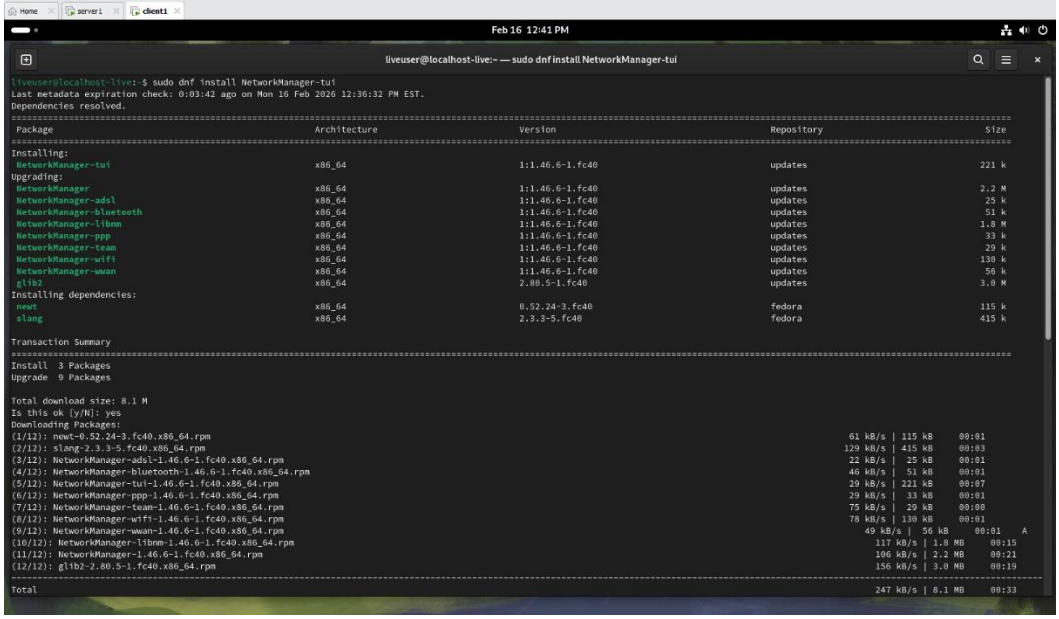

Activities
Terminal
Feb 16 22:48
server1@localhost:~ — nmtui
NetworkManager TUI
Please select an option
Edit a connection
Activate a connection
Set system hostname
Radio
Quit
<OK>

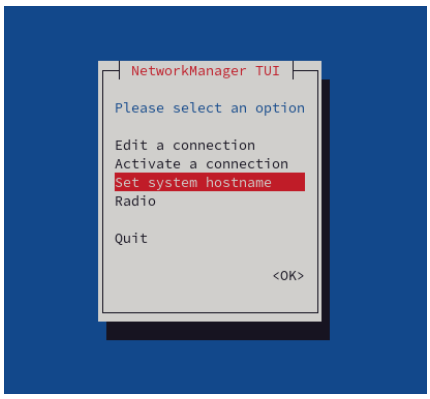

Activities
Terminal
Feb 16 22:49
server1@localhost:~ — nmtui
NetworkManager TUI
Please select an option
Set Hostname
Hostname mlb-dc1-centos7.ndm.lk
<Cancel> <OK>
<OK>

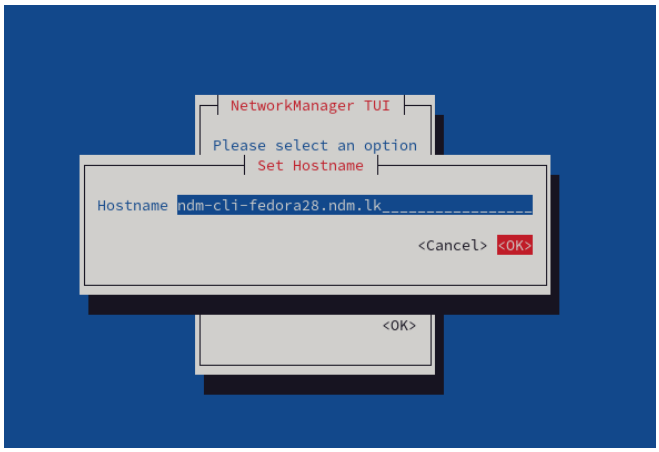
Creating the Client – Specifications



Client-Hostname (Under Network Settings) -nmtui



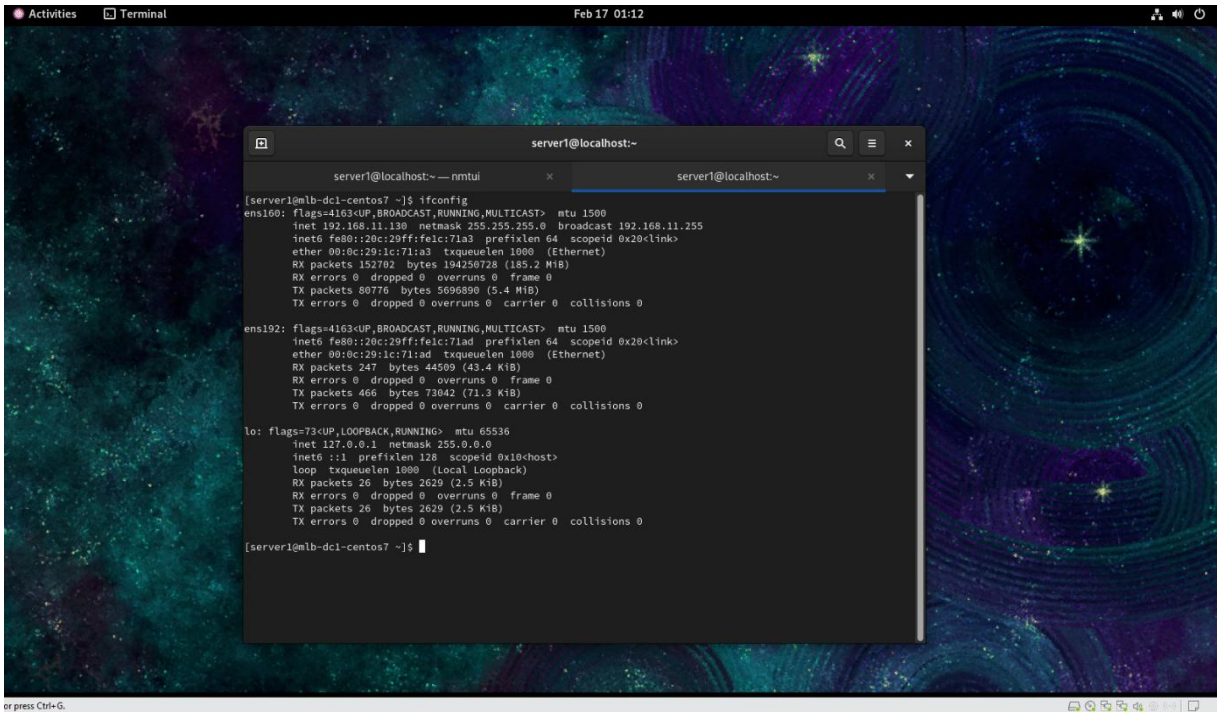




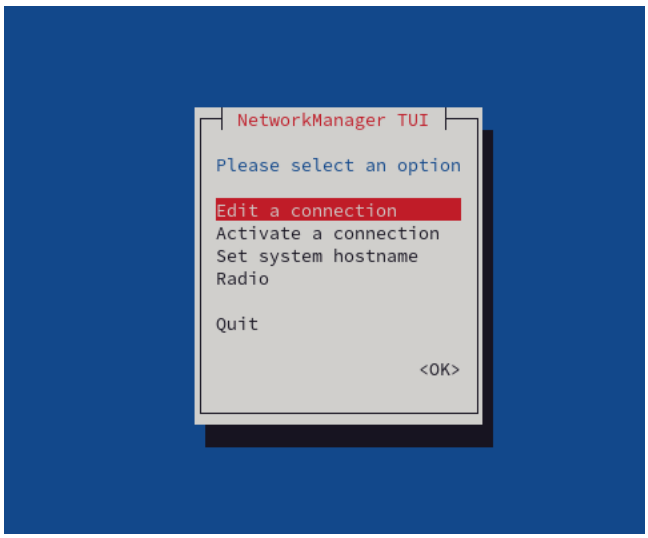
Network Configuration

6.1 Server

Use the command **ifconfig** to list down all the installed Network adapters of the VM. Identify the adapters used by NAT and VMnet2.

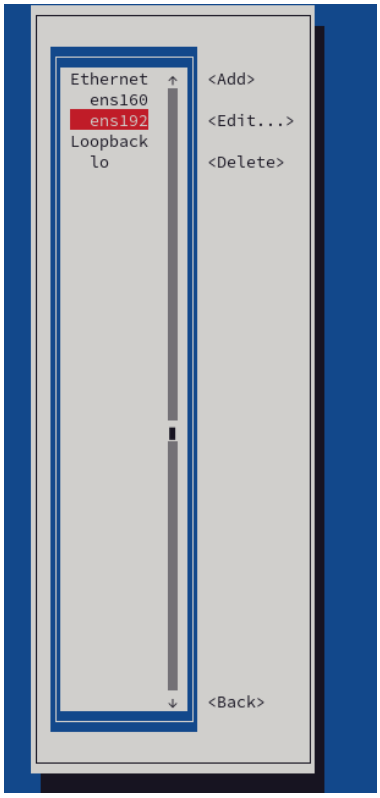


6.1.2 In the open window select edit connection.

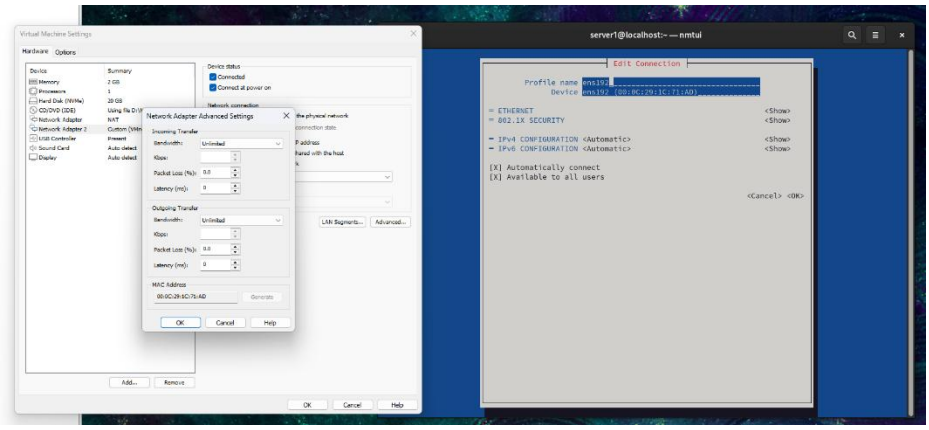


6.1.3 Select the adapter used in VMnet2; select edit.

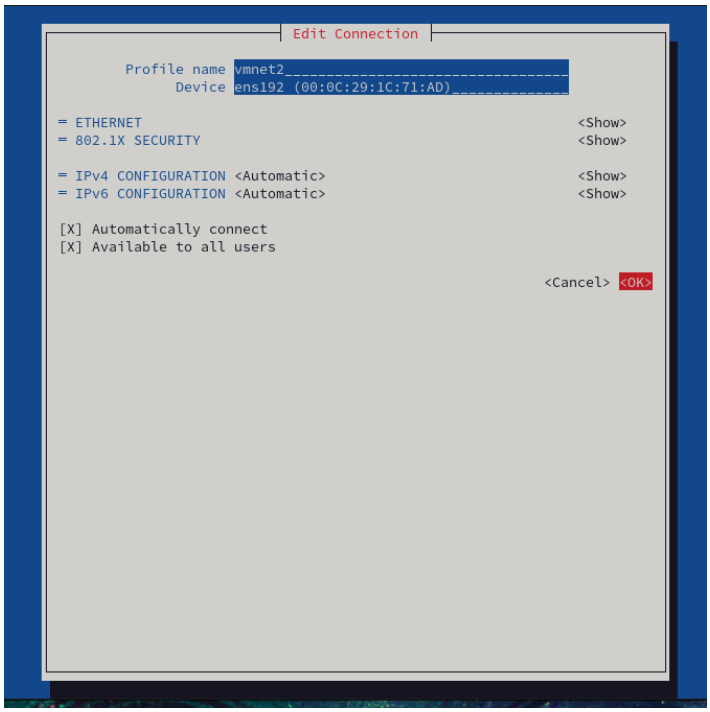
Step 1



Step 2



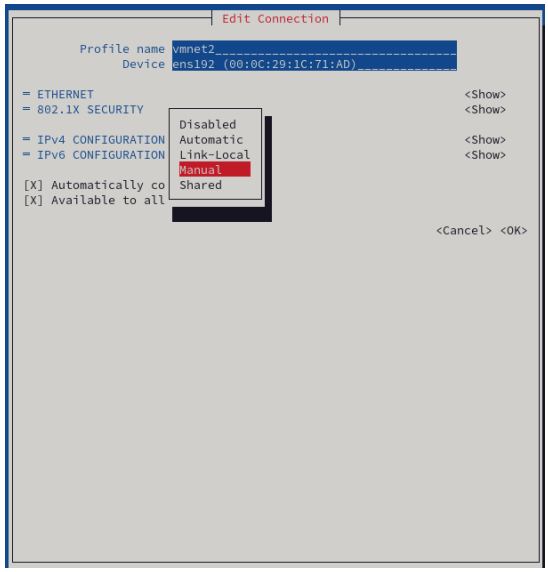
Step 3



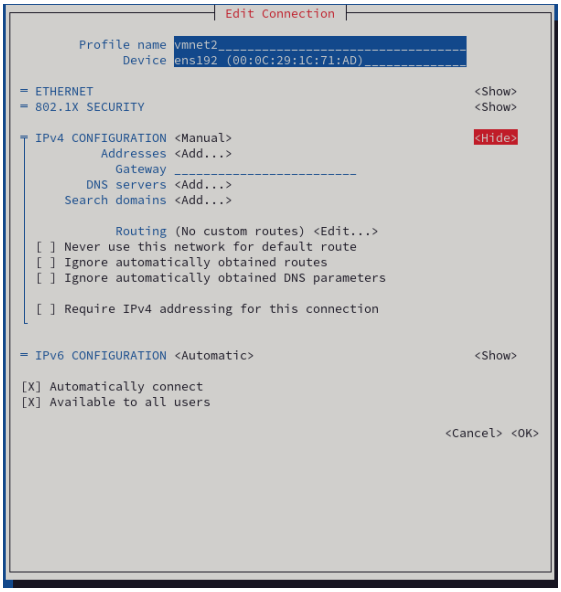
6.1.4 Navigate to IPv4 settings and select the type as Manual and enter the IP configurations given below. IP address: - 10.0.1.2/24 Gateway: - 10.0.1.1

Step 1

Step 2



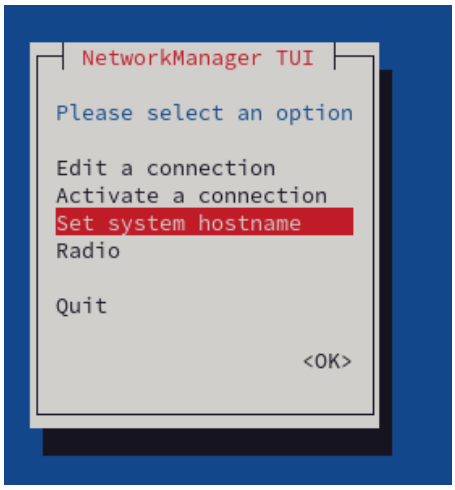
Step 3



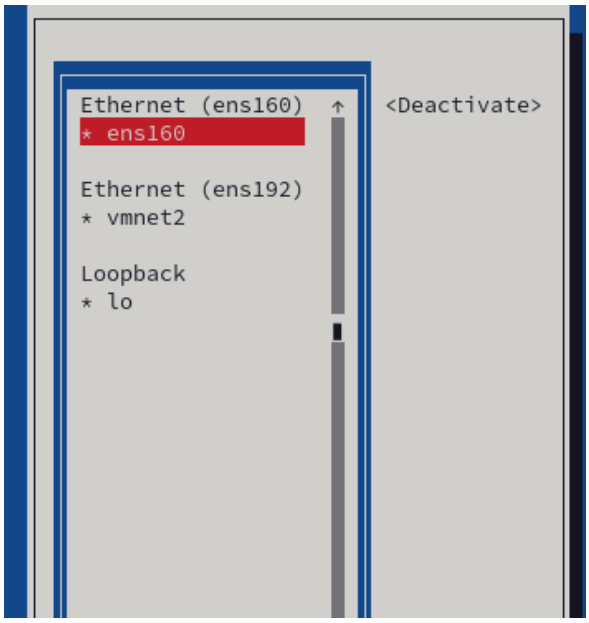
Step 4

6.1.5 Connection Activation can be done using same command 'nmtui'.

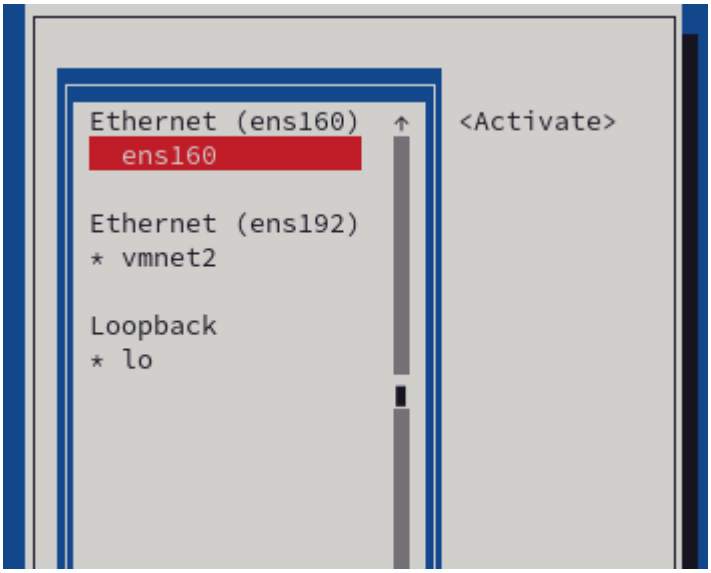
Step 1



Step 2



Step 3

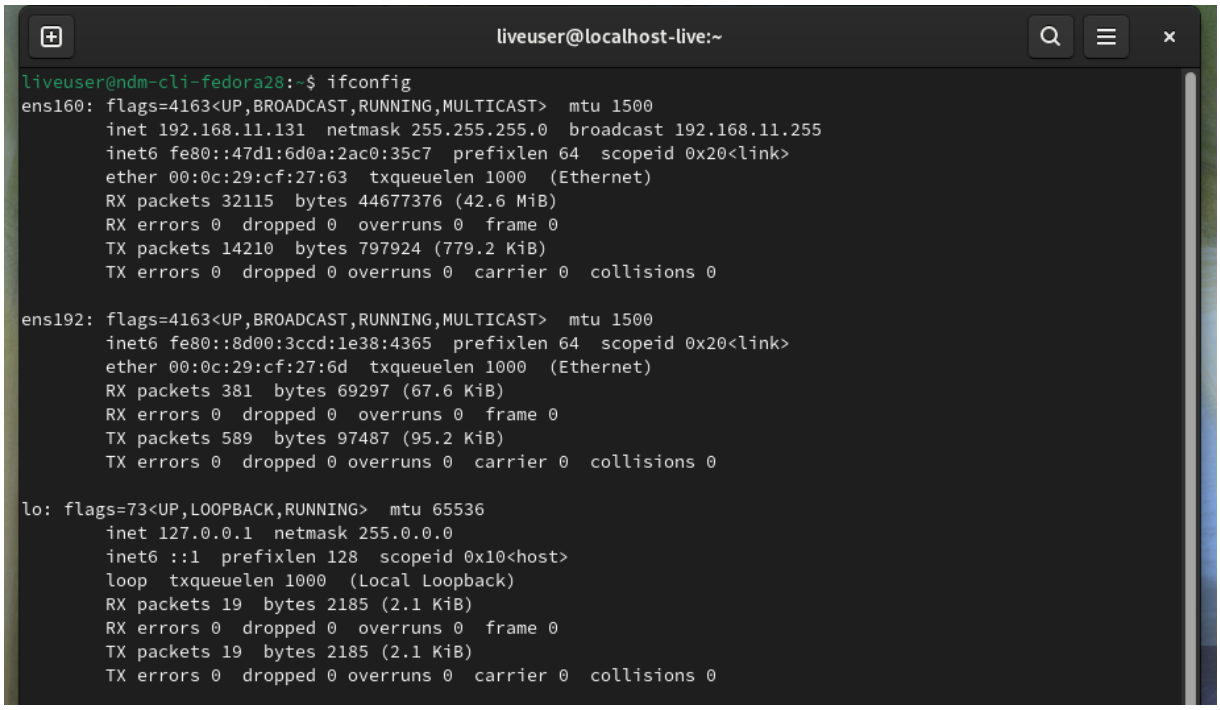


** Mark means activated in connection. - vmnet2 is activated*

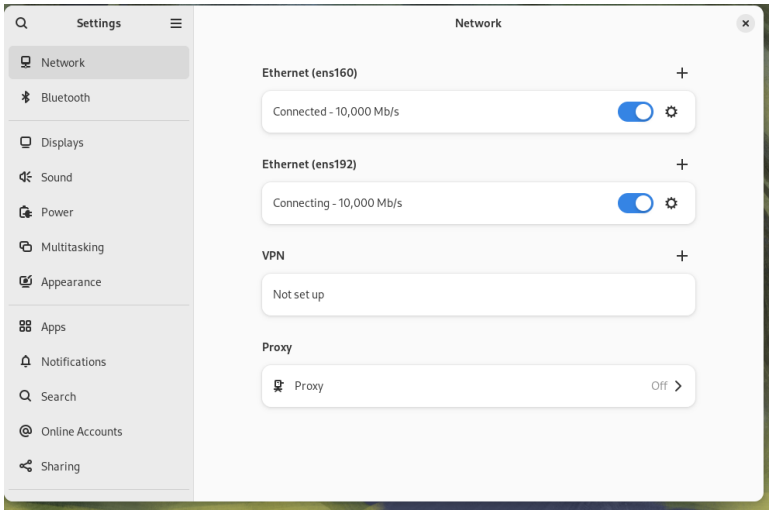
6.2 Client

6.2.1 Configure the Network Adapter connected to VMnet2 using following details and activate the connection. You can use the GUI. IP address: - 10.0.1.3/24 Gateway: - 10.0.1.1

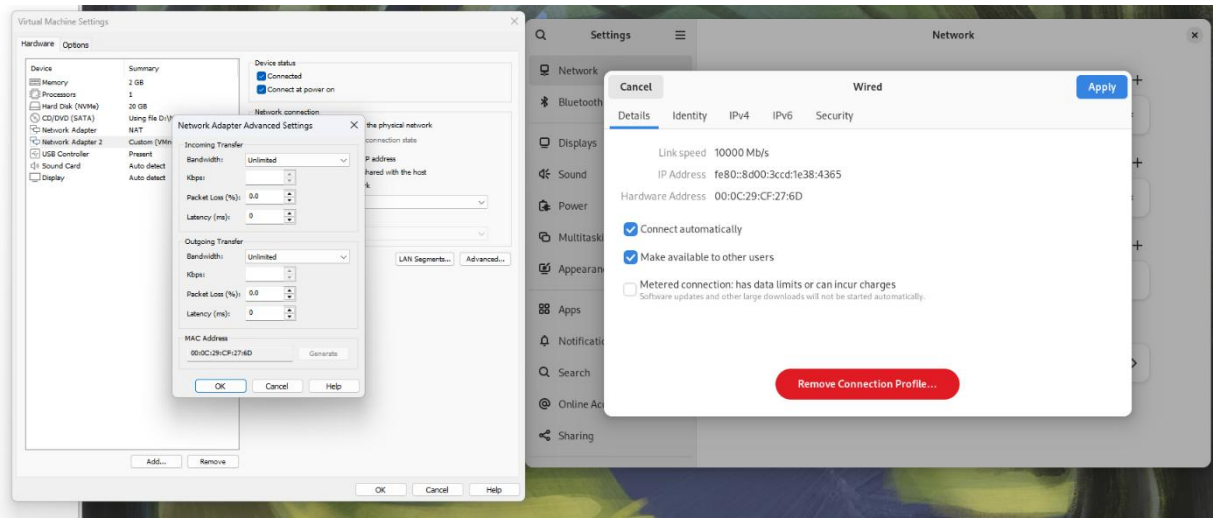
Use the command **ifconfig** to list down all the installed Network adapters of the VM. Identify the adapters used by NAT and VMnet2.



Step 1

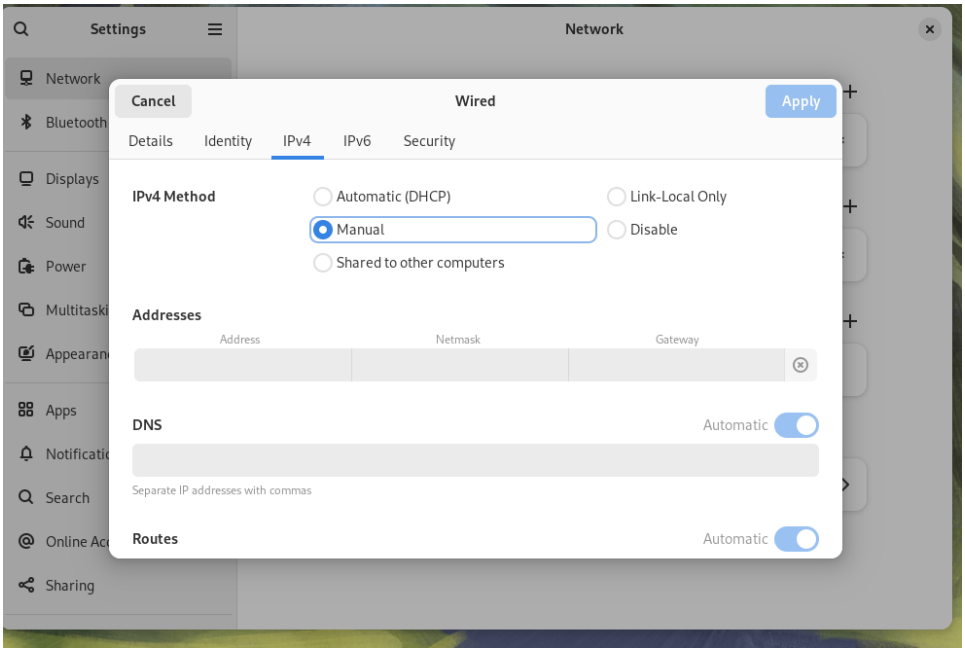


Step 2

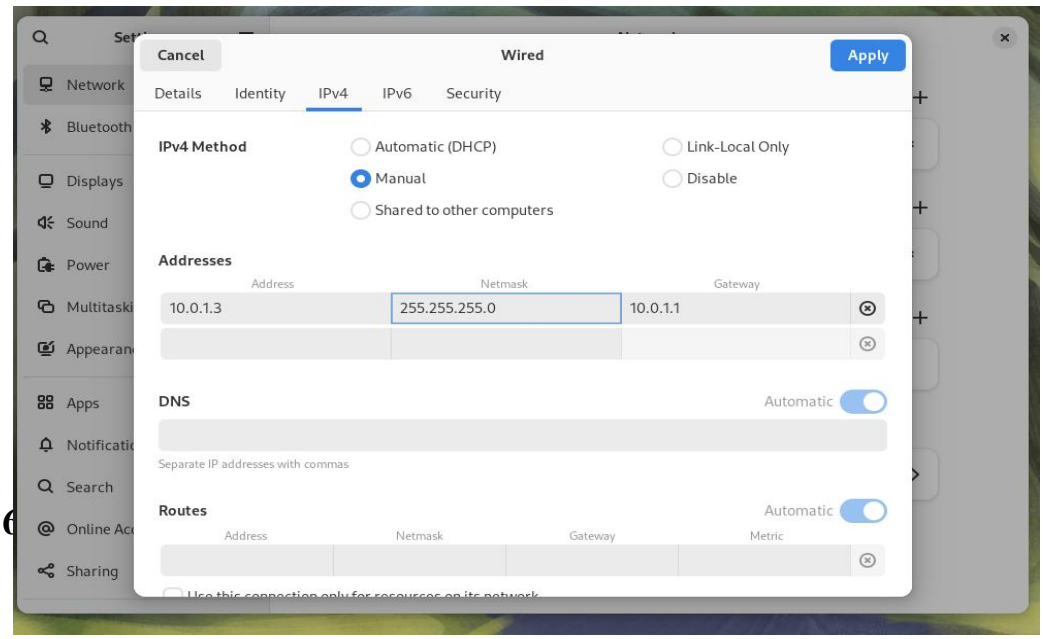


Check mac address

Step 3


Settings
Network
Bluetooth
Displays
Sound
Power
Apps
Search
Sharing
Network
Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
IPv4 Method
Automatic (DHCP)
Link-Local Only
Manual
Disable
Shared to other computers
Addresses
Address
Netmask
Gateway
DNS
Automatic
Separate IP addresses with commas
Routes
Automatic

Step 4


Network
Bluetooth
Displays
Sound
Power
Apps
Search
Sharing
Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
IPv4 Method
Automatic (DHCP)
Link-Local Only
Manual
Disable
Shared to other computers
Addresses
Address
Netmask
Gateway
10.0.1.3
255.255.255.0
10.0.1.1
DNS
Automatic
Separate IP addresses with commas
Routes
Automatic
Address
Netmask
Gateway
Metric

Step 5

```
[server1@mlb-dc1-centos7 ~]$ su -
Password:
[root@mlb-dc1-centos7 ~]# hostname
mlb-dc1-centos7.ndm.lk
[root@mlb-dc1-centos7 ~]#
```

Check host name

Step 6

```
root@mlb-dc1-centos7:~
[server1@mlb-dc1-centos7 ~]$ su -
Password:
[root@mlb-dc1-centos7 ~]# hostname
mlb-dc1-centos7.ndm.lk
[root@mlb-dc1-centos7 ~]# ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=1.21 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=3.94 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=0.177 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=0.131 ms
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=0.134 ms
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=0.175 ms
64 bytes from 10.0.1.2: icmp_seq=7 ttl=64 time=0.176 ms
64 bytes from 10.0.1.2: icmp_seq=8 ttl=64 time=0.168 ms
64 bytes from 10.0.1.2: icmp_seq=9 ttl=64 time=0.178 ms
64 bytes from 10.0.1.2: icmp_seq=10 ttl=64 time=0.146 ms
64 bytes from 10.0.1.2: icmp_seq=11 ttl=64 time=0.094 ms
64 bytes from 10.0.1.2: icmp_seq=12 ttl=64 time=0.106 ms
64 bytes from 10.0.1.2: icmp_seq=13 ttl=64 time=0.137 ms
64 bytes from 10.0.1.2: icmp_seq=14 ttl=64 time=0.116 ms
64 bytes from 10.0.1.2: icmp_seq=15 ttl=64 time=0.095 ms
^C
--- 10.0.1.2 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14314ms
rtt min/avg/max/mdev = 0.094/0.465/3.935/0.965 ms
[root@mlb-dc1-centos7 ~]#
```

6.2.3 Send messages/data packets in :-

- ✓ Server to client
- ✓ Client to server

```
root@mlb-dc1-centos7:~
[root@mlb-dc1-centos7 ~]# hostname
mlb-dc1-centos7.ndm.lk
[root@mlb-dc1-centos7 ~]# ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=1.21 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=3.94 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=0.177 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=0.131 ms
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=0.134 ms
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=0.175 ms
64 bytes from 10.0.1.2: icmp_seq=7 ttl=64 time=0.176 ms
64 bytes from 10.0.1.2: icmp_seq=8 ttl=64 time=0.168 ms
64 bytes from 10.0.1.2: icmp_seq=9 ttl=64 time=0.178 ms
64 bytes from 10.0.1.2: icmp_seq=10 ttl=64 time=0.146 ms
64 bytes from 10.0.1.2: icmp_seq=11 ttl=64 time=0.094 ms
64 bytes from 10.0.1.2: icmp_seq=12 ttl=64 time=0.106 ms
64 bytes from 10.0.1.2: icmp_seq=13 ttl=64 time=0.137 ms
64 bytes from 10.0.1.2: icmp_seq=14 ttl=64 time=0.116 ms
64 bytes from 10.0.1.2: icmp_seq=15 ttl=64 time=0.095 ms
^C
--- 10.0.1.2 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14314ms
rtt min/avg/max/mdev = 0.094/0.465/3.935/0.965 ms
[root@mlb-dc1-centos7 ~]# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=64 time=5.88 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=64 time=1.45 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=64 time=1.18 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=64 time=2.63 ms
64 bytes from 10.0.1.3: icmp_seq=5 ttl=64 time=1.49 ms
64 bytes from 10.0.1.3: icmp_seq=6 ttl=64 time=1.35 ms
64 bytes from 10.0.1.3: icmp_seq=7 ttl=64 time=2.34 ms
64 bytes from 10.0.1.3: icmp_seq=8 ttl=64 time=0.753 ms
64 bytes from 10.0.1.3: icmp_seq=9 ttl=64 time=0.965 ms
64 bytes from 10.0.1.3: icmp_seq=10 ttl=64 time=0.674 ms
64 bytes from 10.0.1.3: icmp_seq=11 ttl=64 time=1.40 ms
^C
--- 10.0.1.3 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10078ms
rtt min/avg/max/mdev = 0.674/1.829/5.883/1.402 ms
[root@mlb-dc1-centos7 ~]#
```